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General Chemistry- Chemistry of Elements

Day 04- 17/03/2022

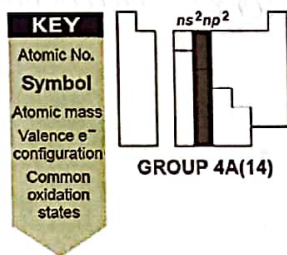
The Carbon Family

14.6 GROUP 4A(14): THE CARBON FAMILY

Bond type

C } Network
Si } Covalent
Ge }

Sn } metallic
Pb }



6	C	12.01 2s ² 2p ² (-4, +4, +2)	
14	Si	28.09 3s ² 3p ² (-4, +4)	
32	Ge	72.61 4s ² 4p ² (+4, +2)	
50	Sn	118.7 5s ² 5p ² (+4, +2)	
82	Pb	207.2 6s ² 6p ² (+4, +2)	
114		(285) 7s ² 7p ²	Observed in experiments at Dubna, Russia, in 1999

Table 14.2 Bond Type and the Melting Process in Groups 3A(13) to 5A(15)

Period	Group 3A(13)				Group 4A(14)				Group 5A(15)				Key:
	Element	Bond Type	Melting Point (°C)	ΔH _{fus} (kJ/mol)	Element	Bond Type	Melting Point (°C)	ΔH _{fus} (kJ/mol)	Element	Bond Type	Melting Point (°C)	ΔH _{fus} (kJ/mol)	
2	B		2180	23.6	C		4100	Very high	N		-210	0.7	Metallic Covalent network Covalent molecule Metal Metalloid Nonmetal
3	Al		660	10.5	Si		1420	50.6	P		44.1	2.5	
4	Ga		30	5.6	Ge		945	36.8	As		816	27.7	
5	In		157	3.3	Sn		232	7.1	Sb		631	20.0	
6	Tl		304	4.3	Pb		327	4.8	Bi		271	10.5	

- Within Group 4A, the large decrease in melting point between the network covalent solids C and Si is due to longer, weaker bonds in the Si structure and the large decrease between Ge and Sn is due to the change from covalent network to metallic bonding.
- Considering horizontal trends, the large increases in melting point and ΔH_{fus} across a period between Al and Si and between Ga and Ge reflect the change from metallic to covalent network bonding.
- Note the unexpected rises in the values for these properties from metallic Al, Ga, and Sn to the network covalent metalloids Si, Ge, and Sb, and note the abrupt drops from the covalent networks of C and Si to the individual molecules of N and P in Group 5A.

Why Graphite is an electric conductor while Diamond is an electric insulator?

In a graphite molecule, one valence electron of each Carbon atom remains free, Thus making graphite a good conductor of electricity. Whereas in Diamond, they have no free mobile electrons. Hence there won't be flow of. That is the reason behind diamond are bad conductors of electricity.

Allotropism: Different Forms of an Element

Allotropes: different crystalline or molecular forms of a substance.

One allotrope is usually more stable than another at a particular pressure and temperature.

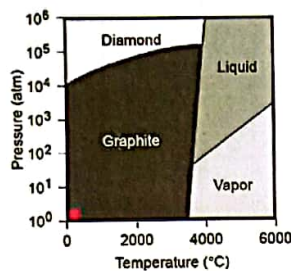
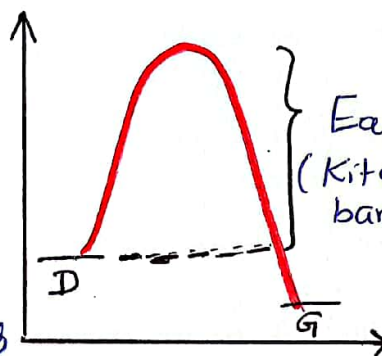


Figure 14.12 Phase diagram of carbon. Graphite is the more stable form of carbon at ordinary conditions (small red circle at extreme lower left). Diamond is more stable at very high pressure.

- Graphite is a black electrical conductor that is soft and "greasy," whereas diamond is a colorless electrical insulator that is extremely hard.
- Graphite is the standard state of carbon, the more stable form at ordinary temperature and pressure.
- Fortunately for jewelry owners, diamond changes to graphite at a negligible rate under normal conditions.



D - Diamond

G - Graphite

* So that's mean Diamonds are kinetically stabilized

* That is the reason where the diamonds are much slowly in decaying into our home into graphite.

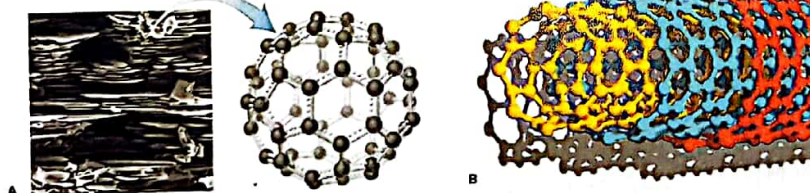


Figure 14.13 Buckyballs and nanotubes. A, Crystals of buckminsterfullerene (C_{60}) next to a ball-and-stick model. The parent of the fullerenes, the "buckyball," is a soccer ball-shaped molecule of 60 carbon atoms. B, Nanotubes are single or, as shown in this colored model, concentric graphite-like tubes (shown here without the usual fullerene ends; see the Gallery, p. 398).

Tin (Sn) has two allotropes

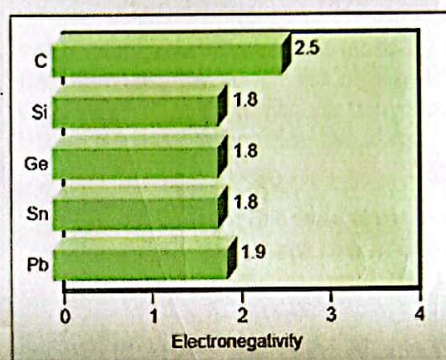
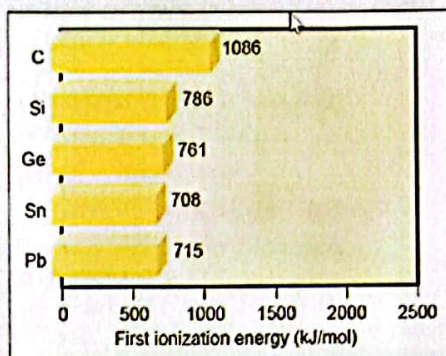
- 650 - 1000
- White tin is stable at room temperature and above, whereas gray tin is the more stable form below 13°C.
 - When white tin is kept for long periods at a low temperature, some converts to microcrystals of gray tin.
 - The random formation and growth of these regions of gray tin, which has a different crystal structure, weaken the metal and make it crumble.

Atomic Properties

Atomic radius (pm)		Ionic radius (pm)
C 77		
Si 118		
Ge 122		
Sn 140		Sn ²⁺ 118
Pb 146		Pb ²⁺ 119

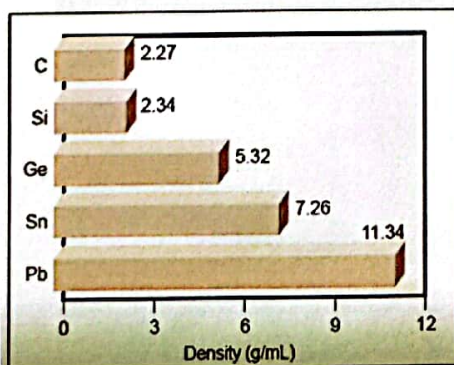
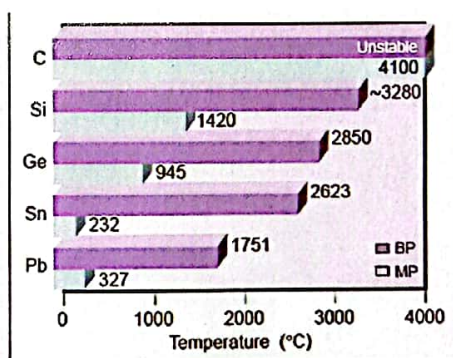
Group electron configuration is ns^2np^2 . Down the group, the number of oxidation states decreases, and the lower (+2) state becomes more common.

Down the group, size increases. Because transition and inner transition elements intervene, IE and EN do not decrease smoothly.



Physical Properties

Trends in properties, such as decreasing hardness and melting point, are due to changes in types of bonding within the solid: network covalent in C, Si, and Ge; metallic in Sn and Pb (*see text*).



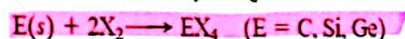
Down the group, density increases because of several factors, including differences in crystal packing.

Some Reactions and Compounds

Important Reactions

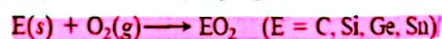
Reactions 1 and 2 pertain to all the elements (E); reactions 3 to 7 concern industrial uses for compounds of C and Si.

1. The elements are oxidized by halogens:

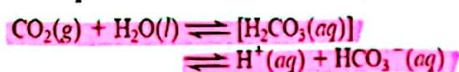


The +2 halides are more stable for tin and lead, SnX_2 and PbX_2 .

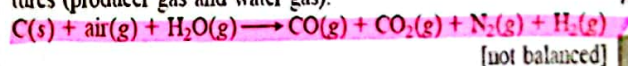
2. The elements are oxidized by O_2 :



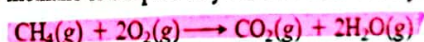
Pb forms the +2 oxide, PbO . Oxides become more basic down the group. The reaction of CO_2 and H_2O provides the weak acidity of natural unpolluted waters:



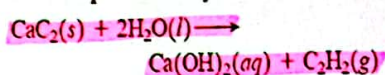
3. Air and steam passed over hot coke produce gaseous fuel mixtures (producer gas and water gas):



4. Hydrocarbons react with O_2 to form CO_2 and H_2O . The reaction for methane is adapted to yield heat or electricity:

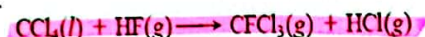


5. Certain metal carbides react with water to produce acetylene:



The gas is used to make other organic compounds and as a fuel in welding (see photo).

6. Freons (chlorofluorocarbons) are formed by fluorinating carbon tetrachloride:



Production of trichlorofluoromethane (Freon-11), the most widely used refrigerant in the world, is being eliminated because of adverse effects on the environment (margin note, p. 589).

7. Silica is reduced to form elemental silicon:



This crude silicon is made ultrapure through zone refining (Gallery, p. 533) for the manufacture of computer chips (see photo).

Computer chip

Oxyacetylene torch

